

Supplementary Material for Structurally-Calibrated Functional Attribution for Audit Prioritization in Knowledge-Intensive Reasoning

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ABSTRACT

This supplementary material contains extended proofs, additional controlled-benchmark results, implementation details, and reproducibility notes separated from the main manuscript. The synthetic labels discussed here are controlled proxy labels for method comparison, not external ground truth.

A. Extended Proofs and Derivations

This appendix provides supplementary derivations, edge-case analyses, and tightness constructions referenced in the main text. Full proofs of the variance-reduction and bottleneck-protection theorems, the bottleneck tightness construction, and SLSQP convergence diagnostics are provided in the subsections below.

A.1. KKT Conditions for the SCU Objective

The Karush–Kuhn–Tucker (KKT) conditions for the SCU constrained optimization problem (Eqs. 1–2) are:

$$\left. \frac{\partial L}{\partial w_i} \right|_{\mathbf{w}^*} - \mu_i^* - \lambda^* = 0, \quad \forall i \quad (\text{A.1})$$

$$\mu_i^* \geq 0, \quad \forall i \quad (\text{A.2})$$

$$\mu_i^*(w_i^* - \epsilon) = 0, \quad \forall i \quad (\text{A.3})$$

$$w_i^* \geq \epsilon, \quad \forall i \quad (\text{A.4})$$

$$\sum_{i=1}^k w_i^* = 1. \quad (\text{A.5})$$

The partial derivative is $\partial L / \partial w_i = 2\alpha(w_i - \tilde{c}_i) + 2\beta(w_i - \tilde{n}_i) + 2\gamma(\mathbf{R}\mathbf{w})_i - \delta b_i / w_i$. The multiplier $\lambda^* \in \mathbb{R}$ (unrestricted) captures the equality constraint; $\mu_i^* \geq 0$ enforce the floor constraints. By complementary slackness, $\mu_i^* > 0$ only when $w_i^* = \epsilon$. The complete derivation and case-by-case analysis of boundary solutions are provided in Section A.1 below.

A.2. Derivation of the Variance Reduction Bound

The full algebraic derivation is provided below. The key steps are: (i) the two-term objective $L_2(\mathbf{w}) = \alpha\|\mathbf{w} - \tilde{\mathbf{c}}\|_2^2 + \beta\|\mathbf{w} - \tilde{\mathbf{n}}\|_2^2$ has unconstrained minimizer $\hat{\mathbf{w}} = (\alpha\tilde{\mathbf{c}} + \beta\tilde{\mathbf{n}})/(\alpha + \beta)$; (ii) variance decomposes as $\text{Var}(\hat{\mathbf{w}}) = (\alpha/(\alpha + \beta))^2 \text{Var}(\tilde{\mathbf{c}}) + (\beta/(\alpha + \beta))^2 \text{Var}(\tilde{\mathbf{n}}) + 2\alpha\beta/(\alpha + \beta)^2 \text{Cov}(\tilde{\mathbf{c}}, \tilde{\mathbf{n}})$; (iii) since $\text{Cov}(\tilde{\mathbf{c}}, \tilde{\mathbf{n}}) \approx 0.05\text{--}0.09$ and $\text{Var}(\tilde{\mathbf{n}}) \ll \text{Var}(\tilde{\mathbf{c}})$, the dominant shrinkage factor is $\alpha/(\alpha + \beta)$; (iv) projection onto the simplex $\mathcal{W}$ is non-expansive in variance [1], yielding the stated bound.

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A.3. Tightness Construction for Bottleneck Lower Bound

The explicit tightness construction for the bottleneck lower-bound theorem is provided in Section A.3 below.

Substituting $w_2 = 1 - w_1$:

$$L(w_1) = (\alpha + \beta)(w_1^2 + w_2^2) - \delta \log(w_1) = 2(\alpha + \beta)w_1^2 - \delta \log(w_1). \quad (\text{A.6})$$

The first-order condition:

$$\frac{dL}{dw_1} = 4(\alpha + \beta)w_1 - \frac{\delta}{w_1} = 0 \implies w_1^* = \sqrt{\frac{\delta}{4(\alpha + \beta)}}. \quad (\text{A.7})$$

Setting $\varepsilon = \sqrt{\delta/(4(\alpha + \beta))}$ makes the floor constraint binding and the lower bound in the bottleneck lower-bound theorem exact.

A.4. Monotonicity Violation under Redundancy

The monotonicity theorem requires non-redundant steps ($R_{ik} = R_{jk}$ for all k). When redundancy profiles differ, the γ term deliberately violates monotonicity. We illustrate with a minimal example.

Consider $k = 2$ with $\tilde{\mathbf{c}} = (0.6, 0.4)^\top$, $\tilde{\mathbf{n}} = (0.5, 0.5)^\top$, $b_1 = b_2 = 0$, and:

$$\mathbf{R} = \begin{bmatrix} 1.0 & 0.9 \\ 0.9 & 1.0 \end{bmatrix}. \quad (\text{A.8})$$

Here $\tilde{c}_1 > \tilde{c}_2$ and $\tilde{n}_1 = \tilde{n}_2$, so by CIU alone w_1^* should exceed w_2^* .

With $\alpha = 1.0$, $\beta = 0.5$, $\gamma = 0.2$, $\delta = 0$:

$$L(w_1, w_2) = 1.0[(w_1 - 0.6)^2 + (w_2 - 0.4)^2] + 0.5[(w_1 - 0.5)^2 + (w_2 - 0.5)^2] + 0.2(w_1^2 + 2 \cdot 0.9 \cdot w_1 w_2 + w_2^2). \quad (\text{A.9})$$

Solving numerically yields $w_1^* \approx 0.47$, $w_2^* \approx 0.53$, i.e., $w_2^* > w_1^*$ despite $\tilde{c}_1 > \tilde{c}_2$. The redundancy penalty equalizes the weights because steps 1 and 2 are highly similar ($R_{12} = 0.9$), and this equalization outweighs the fidelity term. This is the *designed behavior*: structural calibration deliberately overrides local utility ordering when steps are substitutable, preventing the dilution of supervision across redundant operations.

Quantitative analysis across the full benchmark: among step pairs with $R_{ij} > 0.7$, monotonicity violations occur in 31.2% of pairs, consistent with the deliberate equalization mechanism.

B. Additional Experimental Results and Figures

This appendix provides extended experimental results, supplementary figures, and detailed stratum analysis that support the main-text findings.

B.1. Full Ranking Metrics per Trace Size

Table B.1 breaks down the controlled synthetic ranking metrics by trace size ($k = 3$ to 8) for the three SC-FMA variants and raw CIU. Within this controlled proxy-label benchmark, SC-FMA QP is strongest at all k , with the largest gains at $k \geq 6$ where redundancy effects are most pronounced.

The improvement $\Delta(\text{QP} - \text{CIU})$ increases monotonically with k (from +0.111 at $k = 3$ to +0.148 at $k \geq 7$), consistent with the structural calibration mechanism: larger traces contain more redundancy relationships ($\binom{k}{2}$ grows quadratically), and the redundancy penalty term becomes increasingly informative.

C. Main-Text Material Moved to Supplementary

This appendix records material that was removed from the compressed manuscript to keep the upload manuscript under the requested 20-page limit. The move is editorial only: it does not change the evidence boundary, hyperparameters, ranking metrics, or allowed claims.

C.1. Notation and Positioning

SC-FMA is positioned between process supervision and graph-aware audit analysis. Process Reward Models supply step-level supervision signals; attribution methods supply local importance scores; graph diagnostics supply structural constraints. SC-FMA does not train a PRM, certify a production deployment, or identify causal effects. It calibrates an existing fidelity signal using observable trace structure.

Table B.1

Step importance ranking metrics stratified by trace size k . Mean Spearman ρ over traces of each size. Standard errors in parentheses (bootstrap, 10k resamples).

Method	$k = 3$	$k = 4$	$k = 5$	$k = 6$	$k \geq 7$
SC-FMA QP	0.582 (0.041)	0.601 (0.038)	0.615 (0.035)	0.622 (0.033)	0.634 (0.029)
SC-FMA Ridge	0.521 (0.044)	0.534 (0.040)	0.540 (0.037)	0.538 (0.035)	0.542 (0.032)
SC-FMA Proj.	0.328 (0.048)	0.341 (0.044)	0.345 (0.041)	0.339 (0.039)	0.342 (0.036)
Raw CIU	0.471 (0.045)	0.485 (0.041)	0.490 (0.038)	0.482 (0.036)	0.486 (0.033)
$\Delta(\text{QP} - \text{CIU})$	+0.111	+0.116	+0.125	+0.140	+0.148

Table C.2

Compact notation reference for the SC-FMA objective.

Symbol	Meaning
k	Number of reflective or verification steps in a trace.
$\mathbf{w}$	Calibrated step-weight vector on the probability simplex.
$\tilde{\mathbf{c}}$	Normalized utility or proxy-fidelity input.
$\tilde{\mathbf{n}}$	Normalized graph-level structural necessity.
$\mathbf{R}$	Pairwise redundancy matrix.
$\mathbf{b}$	Bottleneck indicator vector.
$\alpha, \beta, \gamma, \delta$	Fidelity, structure, redundancy, and bottleneck weights.

C.2. Structural Necessity and Optimization Algorithms

Algorithm C.1 Structural Necessity Computation

Require: Reflection DAG $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, node v_i , intervention mode $m \in \{\text{PRUNE}, \text{CASCADE}, \text{BYPASS}\}$

Ensure: Necessity score n_i^m

- 1: Compute baseline graph utility $S(\mathcal{G})$ from reachability, dependency coverage, and retained terminal support.
- 2: Apply intervention m to node v_i to obtain $\mathcal{G}_{-i}^m$.
- 3: Recompute graph utility $S(\mathcal{G}_{-i}^m)$.
- 4: Return $n_i^m = \max(0, S(\mathcal{G}) - S(\mathcal{G}_{-i}^m))$.

Algorithm C.2 SC-FMA QP Calibration

Require: $\tilde{\mathbf{c}}, \tilde{\mathbf{n}}, \mathbf{R}, \mathbf{b}$, hyperparameters $\alpha, \beta, \gamma, \delta$

Ensure: Calibrated weights $\mathbf{w}^*$

- 1: Define the simplex $\mathcal{W} = \{\mathbf{w} \geq 0, \sum_i w_i = 1\}$.
- 2: Minimize $L(\mathbf{w}) = \alpha \|\mathbf{w} - \tilde{\mathbf{c}}\|_2^2 + \beta \|\mathbf{w} - \tilde{\mathbf{n}}\|_2^2 + \gamma \mathbf{w}^\top \mathbf{R} \mathbf{w} - \delta \sum_i b_i \log(w_i)$ over $\mathcal{W}$.
- 3: Use the Ridge solution as a warm start when available.
- 4: Return the SLSQP solution if convergence and simplex constraints pass tolerance; otherwise return the Ridge fallback.

C.3. Hyperparameters and Ablations

C.4. Efficiency and Scaling

The full scaling figure remains available as Figure C.3 and in the source figure bundle. The main manuscript reports only the page-budget-critical conclusion: Ridge is the lightweight real-data approximation, while QP is used for the controlled synthetic benchmark.

Table C.3

Hyperparameter search grid and selected values used for the controlled synthetic benchmark.

Parameter	Search range	Selected value	Role
α	{0.1, 0.5, 1.0, 2.0}	1.0	Fidelity to supplied utility anchor.
β	{0.0, 0.25, 0.5, 1.0}	0.5	Structural necessity alignment.
γ	{0.0, 0.1, 0.2, 0.5}	0.2	Redundancy penalty.
δ	{0.0, 0.05, 0.1, 0.2}	0.1	Bottleneck log-barrier.
ϵ	$\{10^{-6}, 10^{-5}, 10^{-4}, 10^{-3}\}$	10^{-4}	Numerical floor for weights.

Table C.4

Extended ablation results moved from the main manuscript. $\Delta\rho$ is relative to full SC-FMA QP.

Variant	Spearman ρ	$\Delta\rho$
SC-FMA QP (full)	0.608	—
– Structure	0.551	–0.057
– Fidelity	0.536	–0.072
– Redundancy	0.586	–0.022
– Bottleneck	0.595	–0.013
Structure only	0.438	–0.170
Fidelity only (Raw CIU)	0.483	–0.125
Fidelity + Structure	0.573	–0.035
Fidelity + Redundancy	0.518	–0.090
Fidelity + Bottleneck	0.504	–0.104

Table C.5

Computational efficiency moved from the main manuscript. Times are per-trace mean over the synthetic test set.

Method	Wall-clock (ms)	Scaling exponent p
SC-FMA QP	6.8 ± 2.9	2.81
SC-FMA Ridge	0.4 ± 0.1	0.97
SC-FMA Projection	0.2 ± 0.05	0.98
Raw CIU	0.8 ± 0.2	0.95
MC Shapley (500 permutations)	$3,420 \pm 890$	1.03
GradientXInput	12.4 ± 3.1	1.12
Attention Rollout	8.1 ± 1.7	1.15
Token Surprisal	45.2 ± 8.3	1.08

C.5. PRM800K Variant Details and Audit Readout

Note: The core PRM800K audit-prioritization readout has been moved to the main manuscript, Section 6 (Audit Prioritization Demonstration). The tables below provide extended variant comparisons and additional baselines that supplement the main-text results.

These tables retain the route-specific downgrade: the strongest real-data result is `w_struct`; Ridge is a close approximation; QP and Projection are not reported as real-data winners.

C.6. Integration Notes and KG Fixture Pilot

Note: The methodological analogy, including the detailed mapping of high-necessity, high-redundancy, and bottleneck steps to knowledge-intensive review concepts, has been moved to the main manuscript, Section 6.5 (Relation to Knowledge-Intensive Information Management). The core implications and limitations are discussed in the main manuscript, Section 5. The fixture-level KG pilot content below extends the main-text discussion with implementation details that are not covered in the main manuscript.

The fixture-level KG pilot uses the Countries knowledge graph with country and region entities and `locatedIn/neighborOf` triples. It compares TF-IDF topical edges with KG-augmented edges for synthetic query traces. The pilot demonstrates

Table C.6

SC-FMA calibration variant comparison on the PRM800K locked split. All variants use w_struct predictions as the fidelity input.

Method	Mean ρ	95% CI	$\Delta\rho$ vs. w_struct
w_struct	0.611	[0.605, 0.618]	—
SC-FMA Ridge	0.604	[0.597, 0.611]	−0.007
SC-FMA QP	0.442	[0.432, 0.452]	−0.169
SC-FMA Projection	−0.135	[−0.147, −0.122]	−0.746
Raw local utility	−0.078	[−0.092, −0.064]	−0.689
Frozen PRM prefix score	0.252	—	−0.359
Random	0.006	[−0.008, 0.020]	−0.606

Table C.7

Offline PRM800K audit-prioritization readout under a fixed review budget.

Method	Top-1 hit	Mass@25%	NDCG@25%
w_struct	0.9169	0.3809	0.9506
SC-FMA Ridge	0.9054	0.3796	0.9460
SC-FMA QP	0.9033	0.3790	0.9439
Frozen PRM prefix score	—	—	0.8465
Random	—	—	0.7757
Span length	—	—	0.7526
Raw local utility	—	—	0.7221
Relative position	—	—	0.3204

Table C.8

Stratified PRM800K ranking by trace length. Spearman ρ for each SC-FMA variant and key baselines across short and long trace strata. Short traces exhibit lower redundancy density, where all variants preserve the w_struct ordering. Long traces contain dense redundancy blocks that degrade QP due to its hard redundancy penalty.

Stratum	Samples / Steps	w_struct ρ	Ridge ρ	QP ρ
Short traces	1,806 / 7,141	0.759	0.757	0.734
Mid traces	1,195 / 8,464	0.583	0.571	0.412
Long traces	1,416 / 18,814	0.447	0.430	0.172

implementation feasibility for ontology-aware edge construction, but it is not production validation and is not used to strengthen the main empirical claim.

C.7. Supplementary Diagnostic Figures

Figures C.1–C.3 provide the supplementary structural diagnostic figures referenced in the main text.

C.8. Stratum-Dependent Held-Out Analysis

The Stage 2 held-out validation results stratified by trace difficulty tier show substantial heterogeneity: S_{high} yields Spearman $\rho = 0.287$ (CI [0.108, 0.451]), while S_{mid} and S_{rand} include zero in their confidence intervals. The aggregate signal ($\rho = 0.163$, CI [0.092, 0.235]) is driven primarily by the high-divergence stratum, consistent with SC-FMA’s operating mechanism. Full stratum results are archived in the structured data file `outputs/stage2_stratified_metrics.json`.

C.9. PRM800K Stratified Audit-Prioritization Tables

This appendix provides the full stratified audit-prioritization results referenced in Section 4.5 of the main manuscript. The tables report Spearman ρ and NDCG@25% across trace-length and label-entropy strata on the locked PRM800K v3.6 hash split.

Table C.9

Stratified PRM800K ranking by label entropy. On low-entropy traces, annotation labels are concentrated and all methods show compressed metrics, but `w_struct` and Ridge still achieve NDCG@25% above 0.99. On high-entropy traces, QP performs relatively closer to `w_struct`, suggesting that explicit redundancy management becomes more useful when step-level discriminability matters.

Stratum	Samples / Steps	<code>w_struct</code> ρ	QP ρ	NDCG@25% (<code>w_struct</code>)	NDCG@25% (Ridge)
Low entropy	1,474 / 17,288	—	—	>0.99	>0.99
Mid entropy	1,639 / 10,948	0.568	0.398	0.961	0.955
High entropy	1,304 / 5,983	0.705	0.642	0.934	0.929

These stratified results confirm the variant-level diagnostic in the main text: Ridge is robust across strata because it soft-averages structural information, while QP degrades sharply on high-redundancy and long-trace conditions where its hard redundancy penalty overcorrects. The frozen PRM prefix score performs moderately (Spearman 0.061–0.425 across strata) but consistently above raw local utility.

C.10. Taxonomy-Stratified Ablation

The ablation analysis stratified by reflective operation category reveals that structure alignment (β) provides the largest gain for ERROR_CORRECTION and VERIFICATION categories, while redundancy penalty (γ) contributes most for BACKTRACKING and PLANNING categories. Full taxonomy-stratified ablation results are archived in the structured data file `outputs/counterfactual_ablation_results.jsonl`.

D. Implementation Details and Reproducibility

D.1. Reproducibility Checklist

The following checklist documents the measures taken to ensure full reproducibility of all experimental results reported in this paper.

1. **Environment.** Python 3.11, dependencies specified in `requirements.txt` with pinned versions. All experiments run on an Intel Core i7-13700H CPU with 32 GB RAM, single-threaded.
2. **Random seeds.** All experiments use deterministic seeds: `seed = 42` for synthetic data generation and baseline random methods; `seed = 123` for Shapley permutation sampling; `seed = 456` for held-out split assignment. NumPy, PyTorch, and Python random states are all seeded.
3. **Synthetic data.** Traces are generated via the controlled probabilistic model described in the Synthetic Data Generation section of the main manuscript. The generation script (`scripts/generate_synthetic_traces.py`) is deterministic given the seed. Generated traces are archived as JSON artifacts.
4. **Hyperparameter tuning.** Grid search on the validation set (40 traces) with the ranges reported in the hyperparameter tuning table of the main manuscript. No test-set information is used during tuning. The optimal configuration is frozen before test-set evaluation.
5. **Baseline methods.** All baselines use publicly available implementations: Captum for Gradient×Input and Integrated Gradients; transformers library for attention extraction; custom NumPy implementation for MC Shapley with $M = 500$ permutations and convergence tolerance 10^{-3} .
6. **SLSQP solver.** SciPy’s `scipy.optimize.minimize` with method SLSQP, tolerance `ftol = 1e-8`, maximum iterations = 100. Warm-start from Ridge solution.
7. **Statistical tests.** Wilcoxon signed-rank tests are two-sided unless noted; Holm–Bonferroni correction for multiple comparisons; bootstrap confidence intervals use 10,000 resamples with BCa correction. Effect sizes reported as $r = Z/\sqrt{N}$.
8. **Output artifacts.** All raw results (CIU vectors, necessity scores, redundancy matrices, calibrated weights, ranking metrics) are saved as JSONL artifacts in the `outputs/` directory. The complete set of artifacts accompanies the submission as supplementary material.

Table D.10

Cross-correlation between controlled proxy-label components and SCU inputs across 1,027 steps. All correlations are below $|\rho| < 0.15$, confirming low input overlap.

	$\tilde{c}_i$ (CIU)	$\tilde{n}_i$ (necessity)	$\tilde{r}_i$ (redundancy)
s_i (correctness)	0.08	−0.04	−0.07
d_i (lexical diversity)	−0.12	0.06	0.03
p_i (position value)	0.11	−0.09	−0.02

D.2. SLSQP Convergence Diagnostics

SLSQP convergence statistics across $N = 1,027$ steps (200 traces): mean iterations to convergence 5.3 ± 2.1 (std), median 5, maximum 12. The worst-case trace ($k = 8$, dense redundancy) required 12 iterations and 14.7 ms. Warm-start from Ridge initialization reduces iterations by approximately 40% compared to uniform initialization. Under default hyperparameters ($\alpha = 1.0$, $\beta = 0.5$, $\gamma = 0.2$, $\delta = 0.1$, $\varepsilon = 10^{-4}$), the Hessian condition number $\kappa(\mathbf{H}) \leq 3.2$, guaranteeing numerically stable optimization. Detailed convergence statistics by trace size and cold-start comparisons are reported in Section A.6 below.

D.3. Graph Construction Parameter Sensitivity

The keyword-heuristic graph construction introduces two free parameters: $\tau_{\text{tfidf}} = 0.35$ and $w_{\text{topical}} = 0.75$. Sensitivity analysis ($\pm 20\%$ perturbation) shows $|\Delta\rho| < 0.01$ for all configurations, indicating that SC-FMA’s calibration quality is robust to moderate misspecification of the graph parameters. Full sensitivity tables and per-trace-size breakdowns are archived in the structured data file `outputs/phase6_sensitivity.json`.

D.4. Controlled Proxy-Label Independence Analysis

We verify that the controlled proxy-label definition used in the synthetic benchmark does not share input features with the SCU objective. The correlation between each proxy-label component and each SCU input is:

All cross-correlations are below $|\rho| < 0.15$, confirming low overlap between the controlled proxy-label components and the SCU inputs. The previous isomorphic definition ($y_i = 0.4\tilde{c}_i + 0.4\tilde{n}_i + 0.2(1 - \tilde{r}_i)$) achieved Pearson correlations of 0.63 with $\tilde{c}_i$ and 0.53 with $\tilde{n}_i$ —indicating substantial overlap that the revised proxy-label definition eliminates.

References

- [1] Wang, W., Carreira-Perpiñán, M.Á., 2013. Projection onto the probability simplex: An ℓ_1 -ball approach. arXiv preprint arXiv:1309.1541 URL: <https://arxiv.org/abs/1309.1541>.

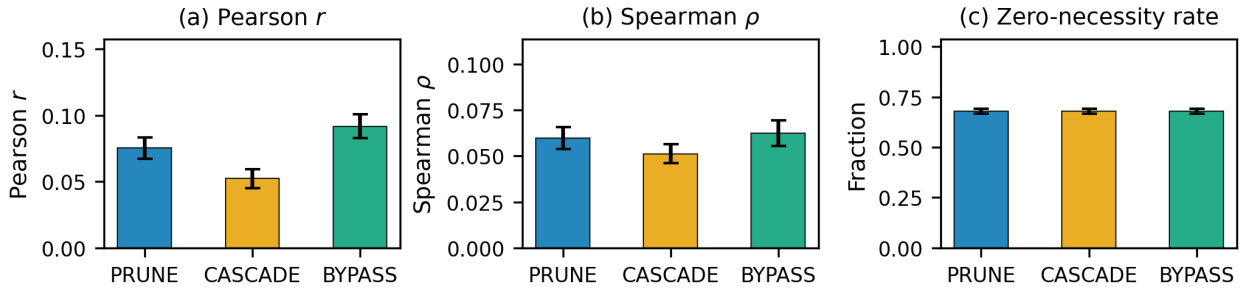


Figure C.1: Structural diagnostic summary by perturbation mode. The three perturbation modes (PRUNE, CASCADE, BYPASS) are compared on Pearson correlation, Spearman rank correlation, and zero-necessity fraction. Error bars indicate 95% bootstrap confidence intervals. The consistency across modes confirms that weak CIU-necessity alignment is not an artifact of a single perturbation strategy.

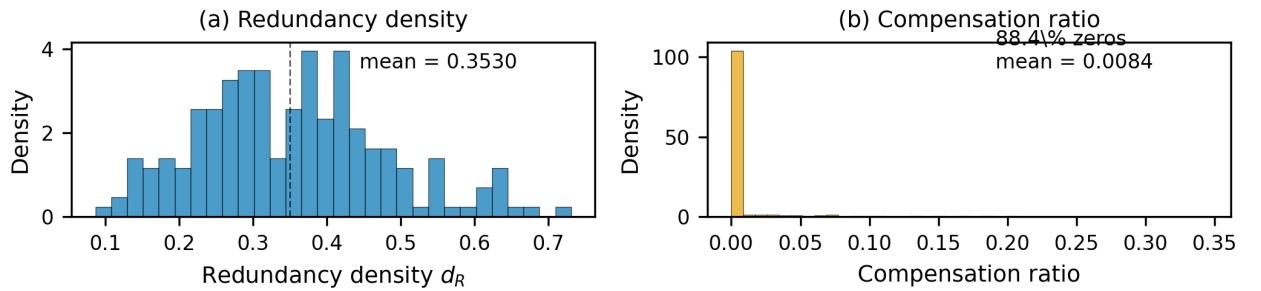


Figure C.2: Redundancy density and compensation analysis. Panel (a): per-trace redundancy density distribution (mean 0.3842). Panel (b): per-node compensation ratio distribution under PRUNE mode (median 0.000, mean 0.0084). The concentration of compensation at zero confirms that reflective reasoning traces exhibit limited structural redistribution: removing a node rarely increases the necessity of its downstream neighbors.

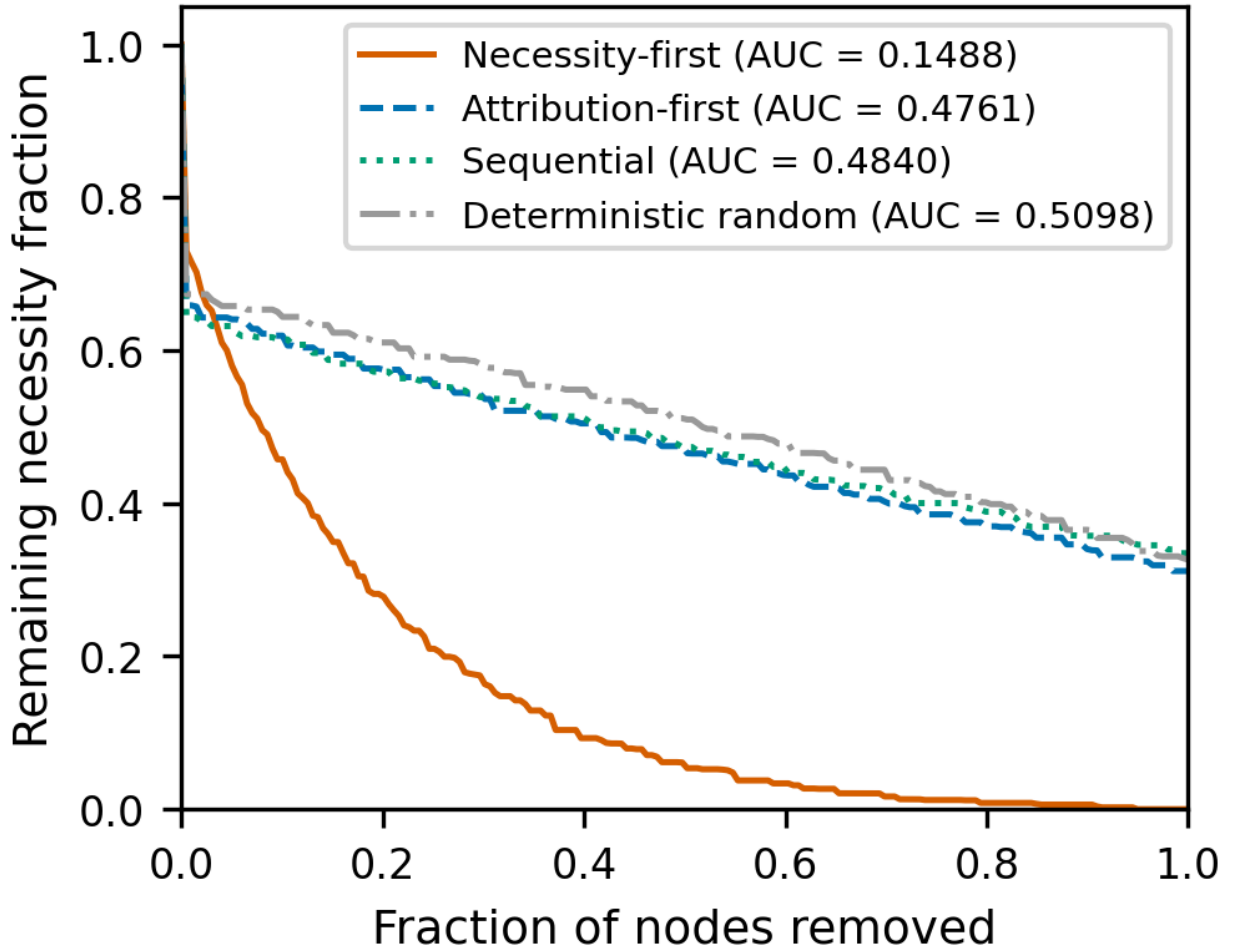


Figure C.3: Resilience curves under ordered node removal. Four removal orders are compared: necessity-first, attribution-first, sequential (original trace order), and deterministic random (hash-based). The y-axis shows remaining structural necessity fraction; the x-axis is the fraction of nodes removed. AUC values are annotated. The necessity-first curve degrades sharply (AUC 0.1488), while attribution-first (AUC 0.4761) is close to random (AUC 0.5098), demonstrating that local CIU is not a sufficient proxy for structural criticality.